XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

Department of Electrical and Electronic Engineering

Xiamen University Malaysia

Quiz 3

COM105 Engineering Physics (I)

6 Questions

Total Marks (100)

27 June 2022 (Tuesday)

9:00 – 10:15 am (One hour fifteen minutes)

INSTRUCTIONS TO CANDIDATES

1. Get your papers and pen ready, all corresponding working steps and answers **MUST be handwritten**. Make sure you have your data cable or have setup any cloud storage service app in your smartphone and PC (iCloud, Dropbox, Onedrive), which allows you to access your phone's photos or scanned images immediately from your PC.
2. This quiz consists of 4 main questions:

Q1:	1	sub-question	Q2:	1	sub-questions
Q3:	2	sub-questions	Q4:	2	sub-questions
Q5:	3	sub-questions	Q6:	5	sub-questions

3. Please follow the requirements of each question, write your **ID, signature, and date** on the top right of **every** pages**.** Indicate the question number i.e., Q1a).
4. Scan your written answer papers with any phone scanner app of your choice, or take picture with the camera app and paste the images to an empty Word document. Make sure the images are clear and readable.
5. Compile all the images and save as **pdf** file. Name the pdf file as **COM105Quiz3_studentID.pdf**.
6. If the file size exceeded 20 MB limit, you may compile multiple **pdf** files according to the question number, as indicated in the file name i.e., **COM105Quiz3_studentID_Q1.pdf**, **COM105Quiz3_studentID_Q2&Q3.pdf**, etc. You are allowed to submit up to 4 **pdf** files in the Moodle Quiz submission tab.
7. You have extra 15 minutes to compile your answer sheet. Submission due at 10:30 am, after which will be considered late and late penalty will be applied.

Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you will be penalized accordingly.

Question 1 (15 marks)

Inside a freezer with temperature of -11°C, a ball made from lead with diameter 5.10 cm is placed at the bottom of a glass bowl, covered entirely by 139 ml of mercury that is filled to the brim of the glass bowl. Determine the amount of mercury that overflows from the bowl it is taken out from the freezer and left to room temperature of 28.0°C? Given that the coefficients of volume expansion for lead, mercury, and glass are 87×10^{-6} °C^{-1}, 180×10^{-6} °C^{-1}, and 27×10^{-6} °C^{-1}, respectively.

Question 2 (10 marks)

A giant 8.90×10^{11} kg space rock lands in the ocean at a speed of 109200 km/h. Assume 1.50% of the asteroid's kinetic energy goes to boiling the ocean water. Calculate the mass of water that will be boiled away by the impact if the initial ocean water temperature is 10.0°C. Given that the specific heat capacities and specific latent heat of vaporisation of ocean water are 4.190 kJ/kg•K and 2260 kJ/kg, respectively.

Question 3 (15 marks)

A wall is constructed using two different materials for the interior and exterior sides: a layer of 5.6 cm thick wood on the outside that can withstand tough weather and a layer of 4.7 cm thick Styrofoam on the inside wall surface for improved thermal insulation. This wall is used in building a small hut situated in a freezing location. A fireplace help keep the interior space warm where the interior wall's surface temperature is 2*x*.0°C, and the exterior surface temperature is -5.0°C, where *x* is the last digit of your ID (*i.e., EEE2109308: 28.0 °C*). Given that the thermal conductivity (k) of the wood and the Styrofoam are 0.080 W/m•K, and 0.027 W/m•K, respectively.

a. Calculate the temperature between the wood and the Styrofoam. **(10 marks)**
b. Calculate the heat flow per square meter through this wall. **(5 marks)**

Question 4 (20 marks)

An enclosed small gas bottle contains 2.20 L of ethane gas (C_2H_6) at temperature of 300 K and atmospheric pressure of 1.013×10^5 Pa. First, the system is warmed up to 610 K while keeping the bottle cap opened. After that, the bottle cap is closed and the system is cooled down to its original temperature. Given that the molar mass of ethane is 30.1 g/mol, find the following:

a. the final pressure of the ethane in the flask. **(10 marks)**
b. the mass (in g) of ethane remained in the flask. **(10 marks)**

Question 5 (20 marks)

A balloon containing 20 L of helium gas at 1.00 atm and a temperature of 15.0°C rises rapidly from ground level to an altitude with 1*x*.0% reduction in atmospheric pressure, where *x* is the last digit of your ID (*i.e.*, *EEE2109308: 18.0 %*). Assume the helium behaves like an ideal gas and there is no heat exchange between the balloon and the surrounding air. At the higher altitude, determine:

a. the volume of the gas. **(5 marks)**
b. the temperature of the gas. **(8 marks)**
c. the change in internal energy of the helium from the initial altitude. **(7 marks)**

Question 6 (20 marks)

A heat engine is fuelled by 1.20 mol of an ideal diatomic gas through the cycle shown in Figure 1.

a. Determine the process of a-b and prove your answer. **(3 marks)**
b. Evaluate and determine the process(es) of the cycle where heat is absorbed and the process(es) where heat is rejected by the gas. **(3 marks)**
c. Calculate the temperature at points a, b, and c. **(5 marks)**
d. Calculate the net heat exchanged with the surroundings and net work done by the engine in one cycle. **(5 marks)**
e. Calculate the thermal efficiency of the engine. **(4 marks)**

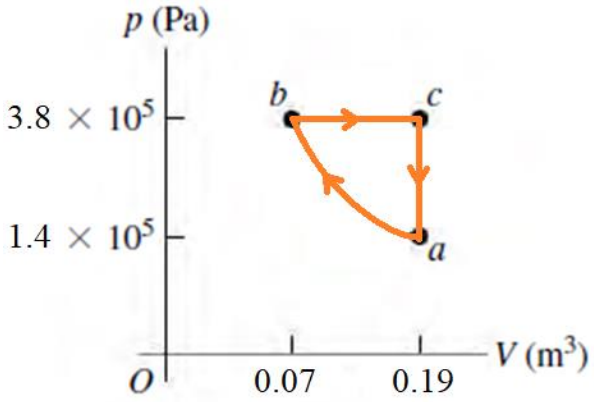


Figure 1